

CO2xNxPhrag: Porewater SO4/Cl

July 2026 Samples

2026-09-17

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-09-14" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "July"
Site = "GCRew"
Project_name = "CO2xNxPhrag"
run_notes = "" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/CO2xNxPhrag_202607_Cl.txt"
raw_file_name_so4 = "Raw Data/CO2xNxPhrag_202607_SO4.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCRew_CO2xNxPhrag_PW_Processed_Cl_SO4_202607.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCRew", "GCRew_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2026.csv"
```

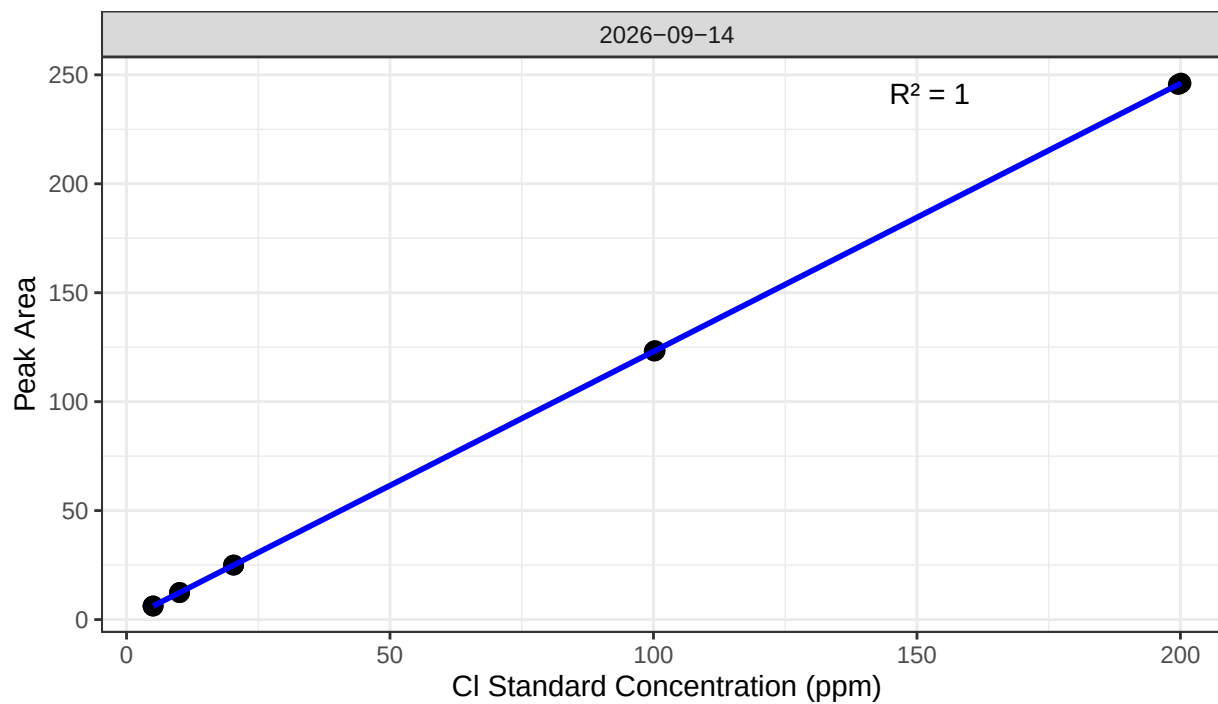
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

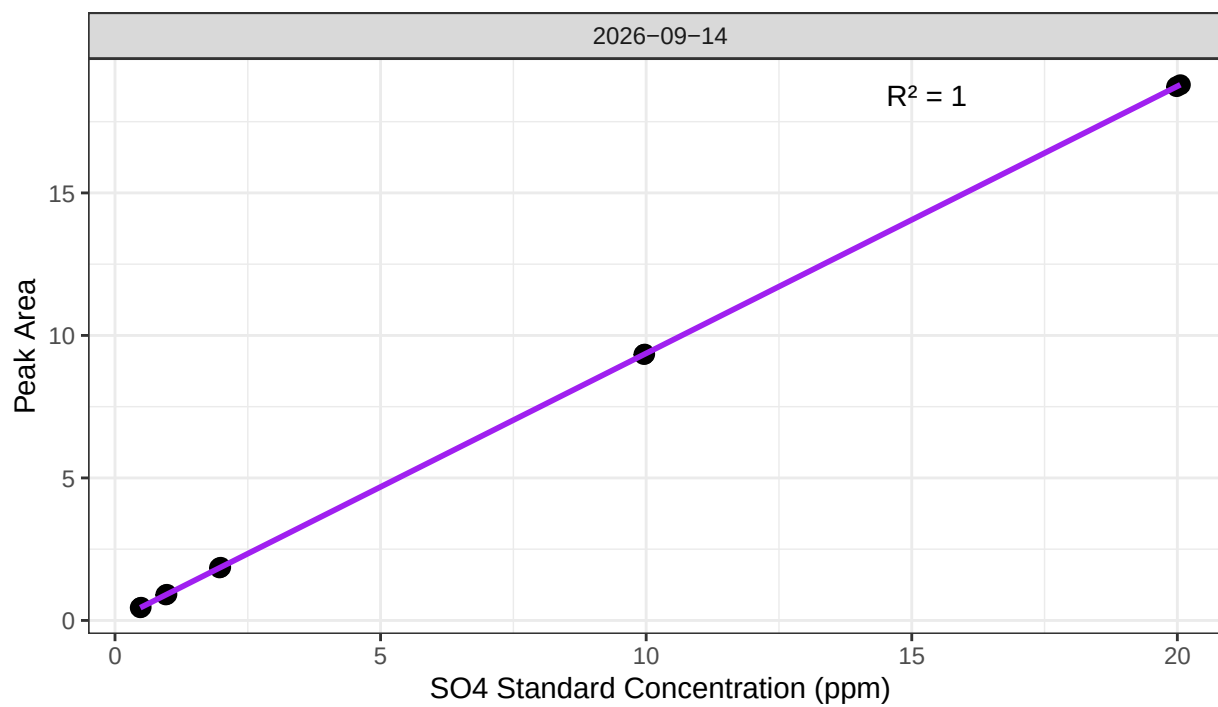
##Import Sample Data

0.2 Assess Standard Curves

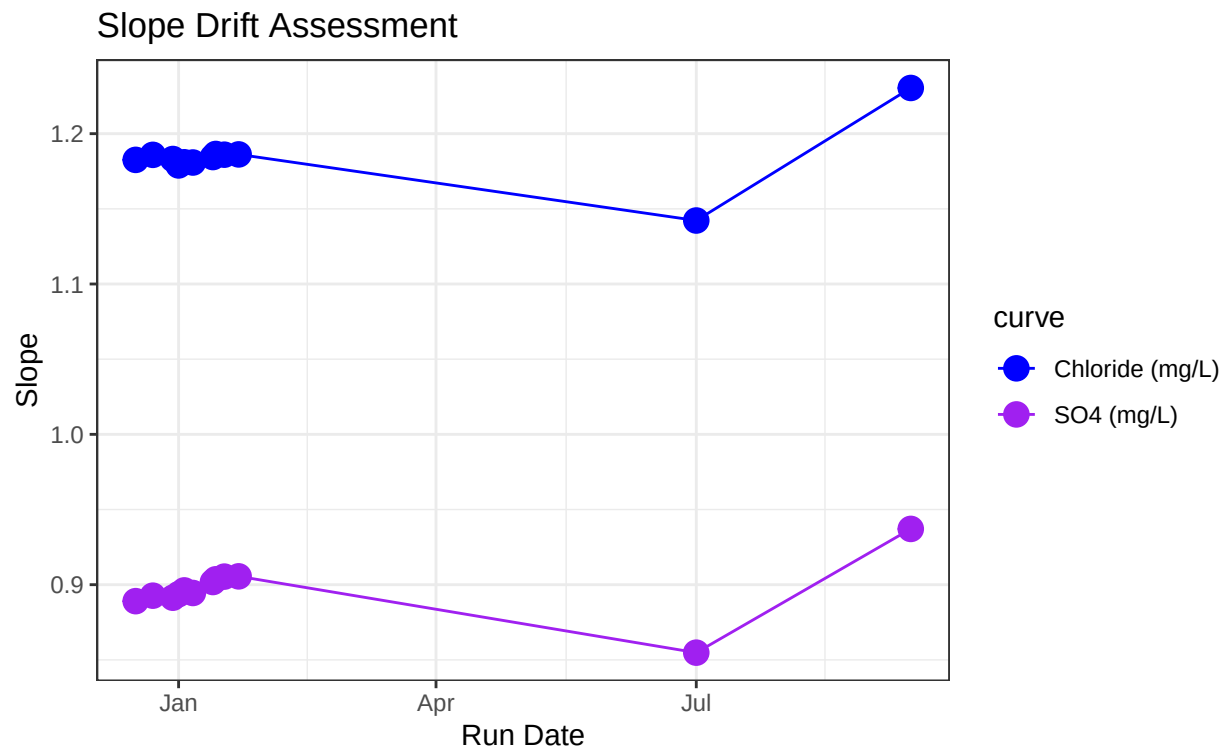
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

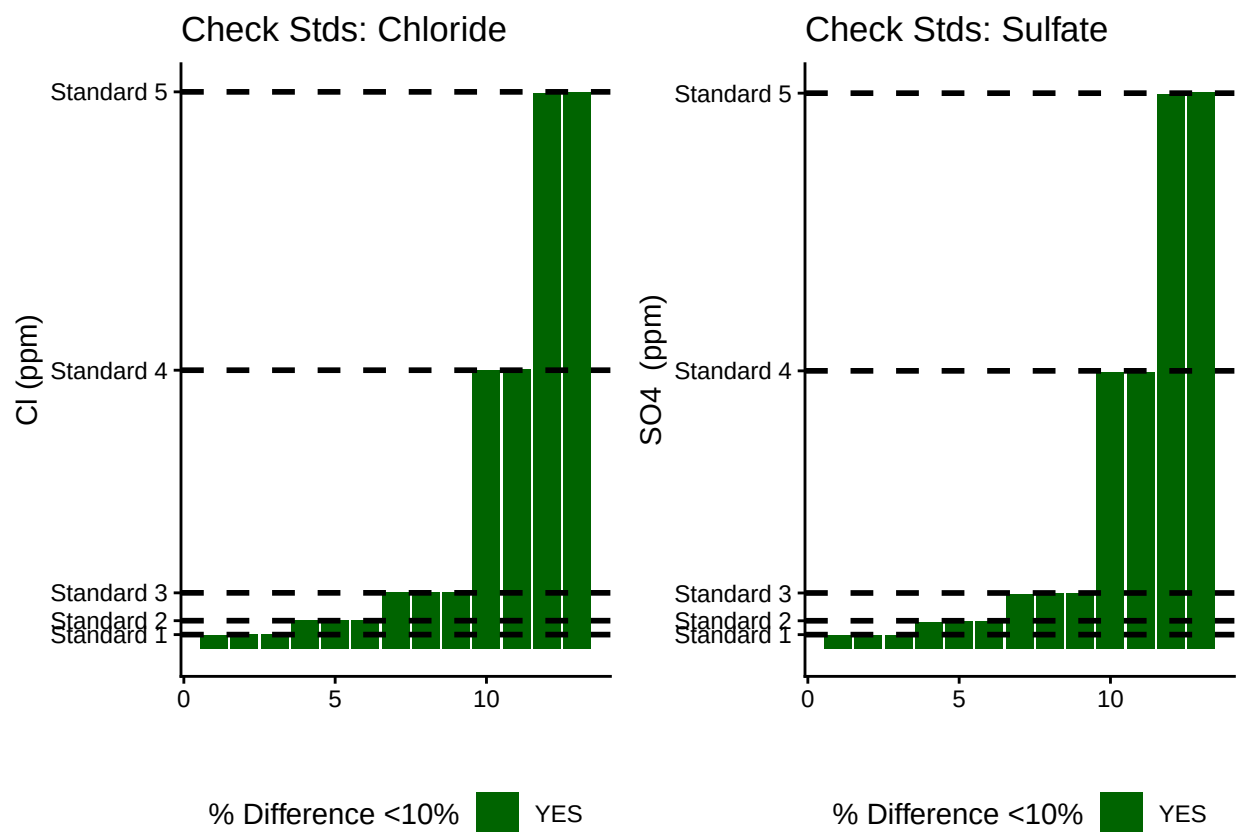
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 5.05 0.0178 0.00352 Chloride Check Standard RSD within Range ~
## 2 Standard 2 10.1 0.0161 0.00160 Chloride Check Standard RSD within Range ~
## 3 Standard 3 20.3 0.00992 0.000488 Chloride Check Standard RSD within Range ~
## 4 Standard 4 100. 0.118 0.00118 Chloride Check Standard RSD within Range ~
## 5 Standard 5 200. 0.358 0.00179 Chloride Check Standard RSD within Range ~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 0.482 0.00895 0.0186 Sulfate Check Standard RSD within Range --
## 2 Standard 2 0.969 0.0109 0.0112 Sulfate Check Standard RSD within Range --
## 3 Standard 3 1.98 0.0128 0.00645 Sulfate Check Standard RSD within Range --
## 4 Standard 4 9.96 0.0103 0.00104 Sulfate Check Standard RSD within Range --
## 5 Standard 5 20.0 0.0499 0.00249 Sulfate Check Standard RSD within Range --
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



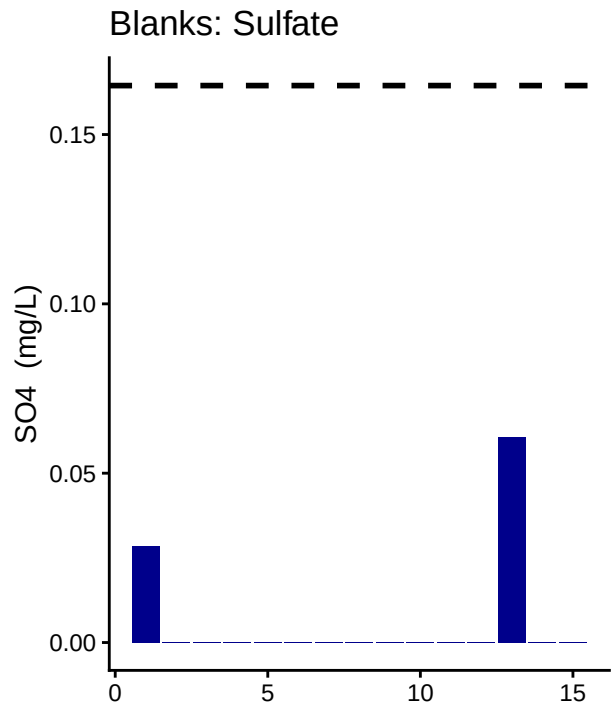
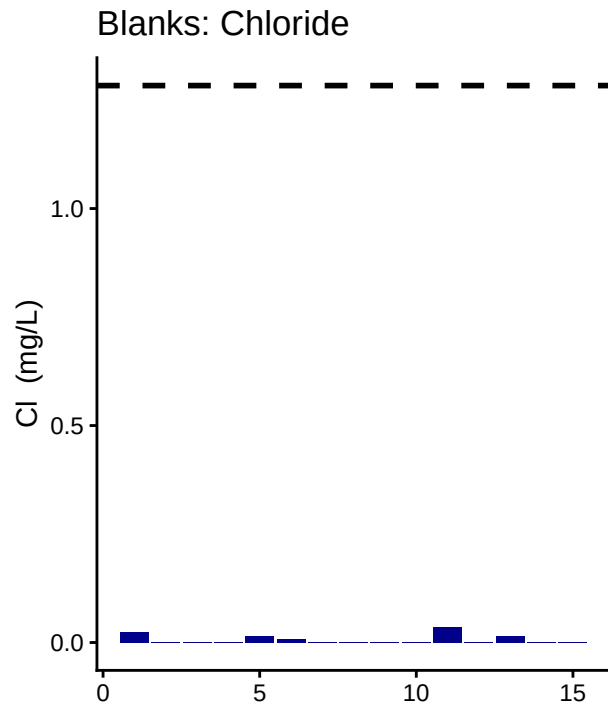
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ Y

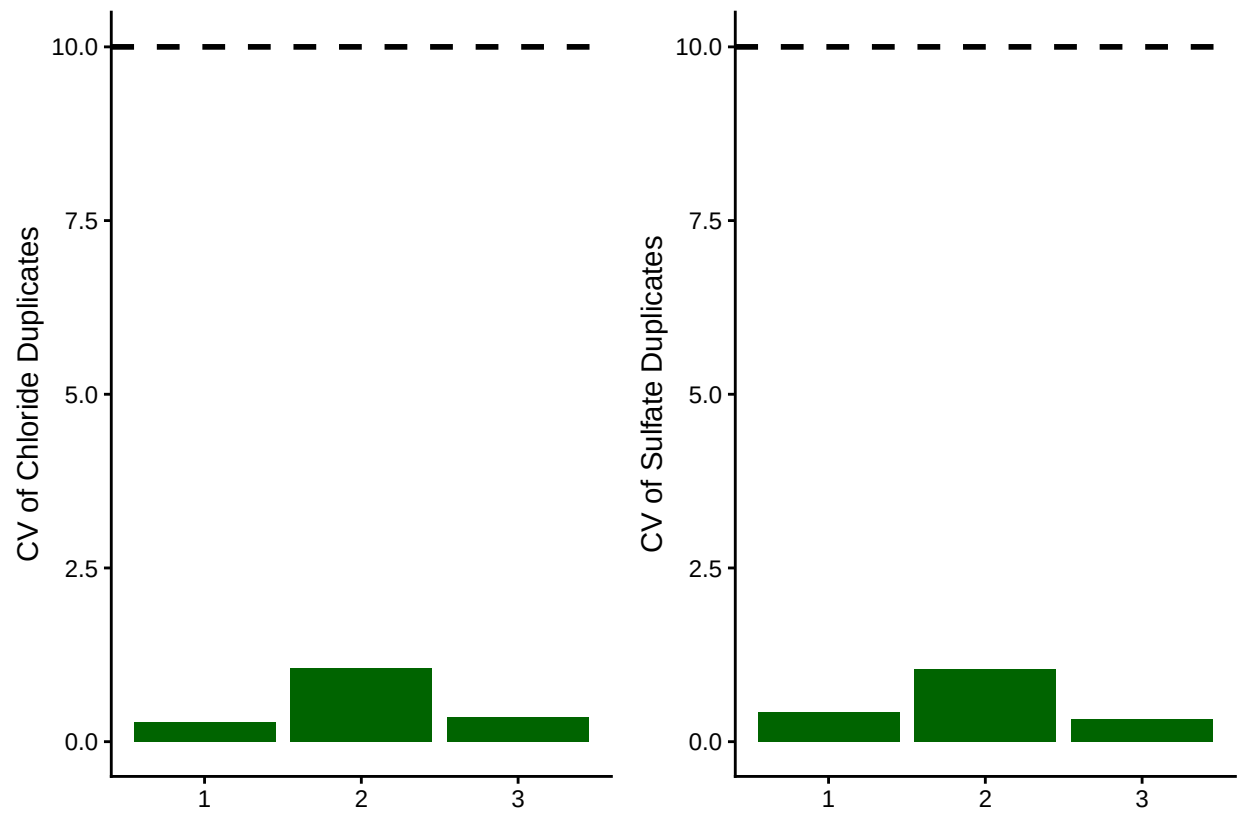
```
## Chloride blanks mean ppm:
```

```
## [1] 0.006633333
```

```
## Sulfate blanks mean ppm:
```

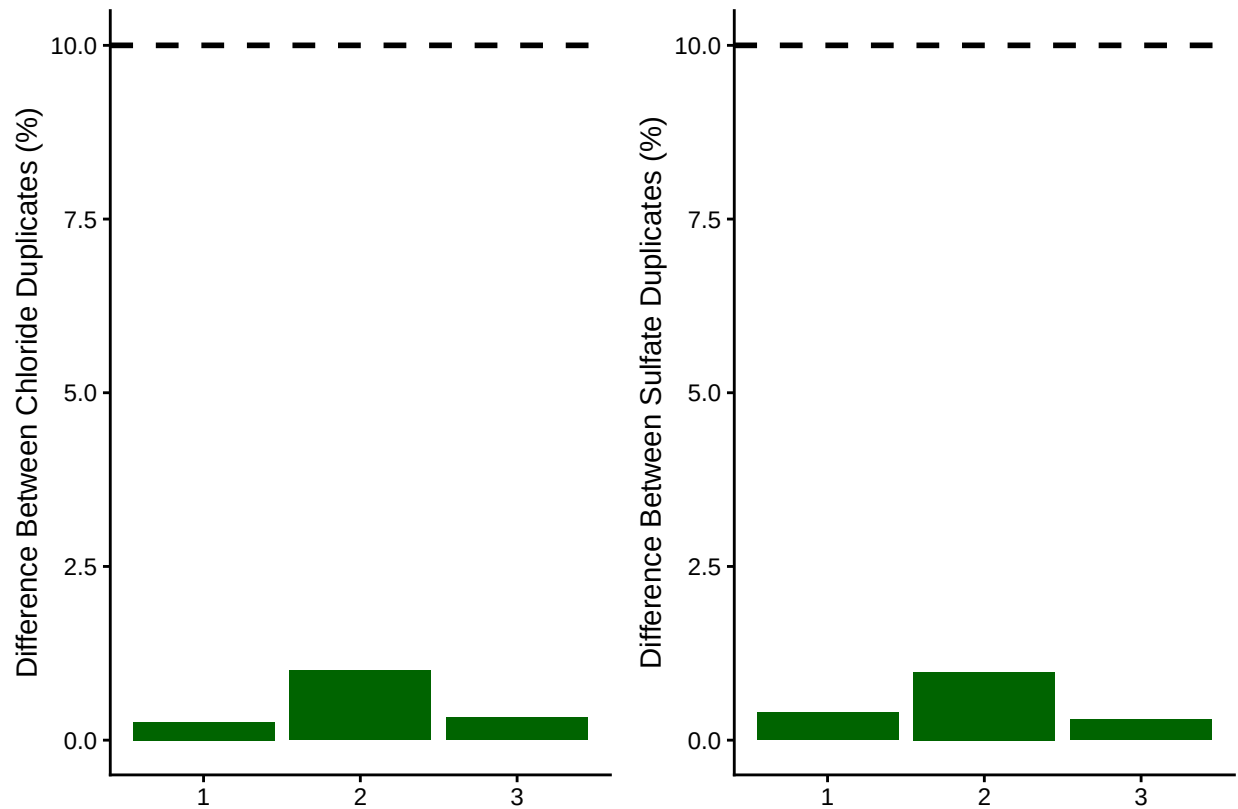
```
## [1] 0.005933333
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

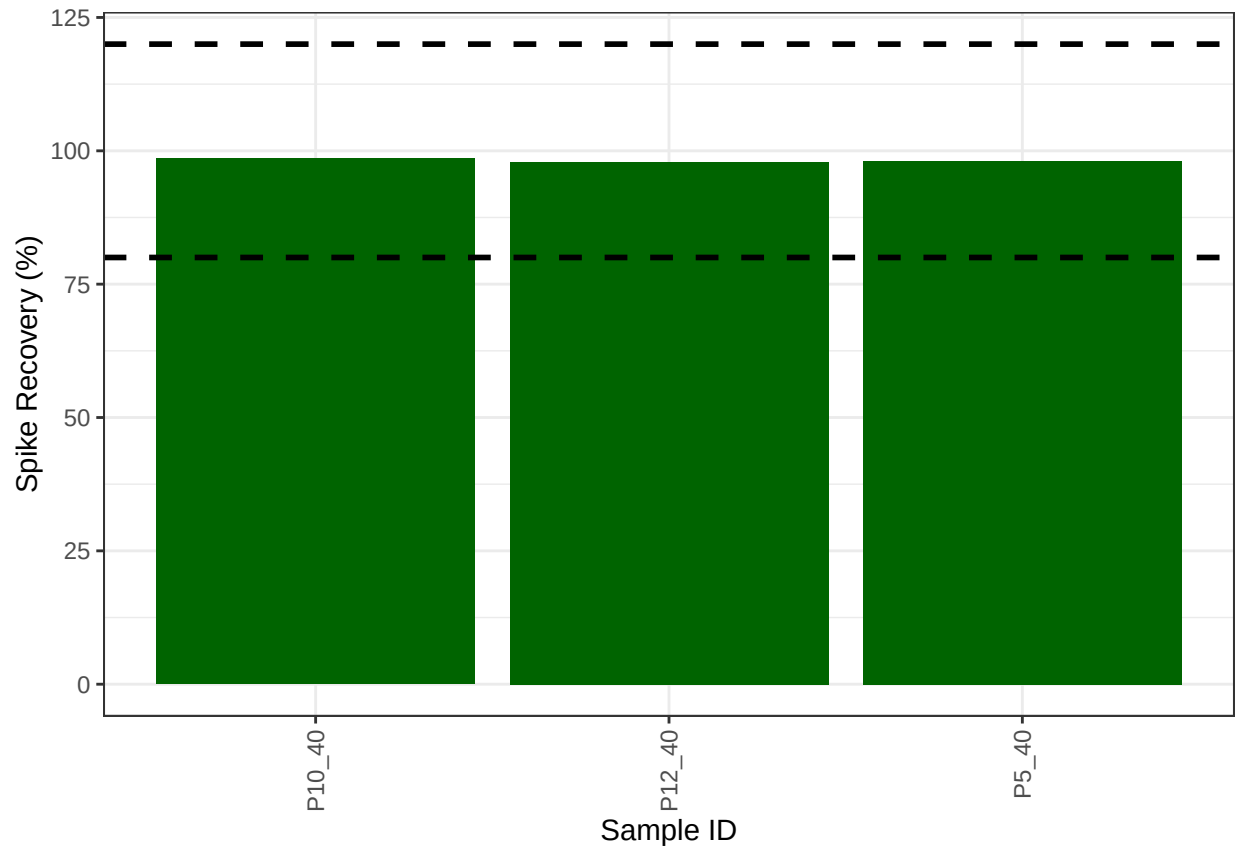
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807 * Cl_ppm) + 0.026) / 1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

# Need to determine dilution factors for your samples
# for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

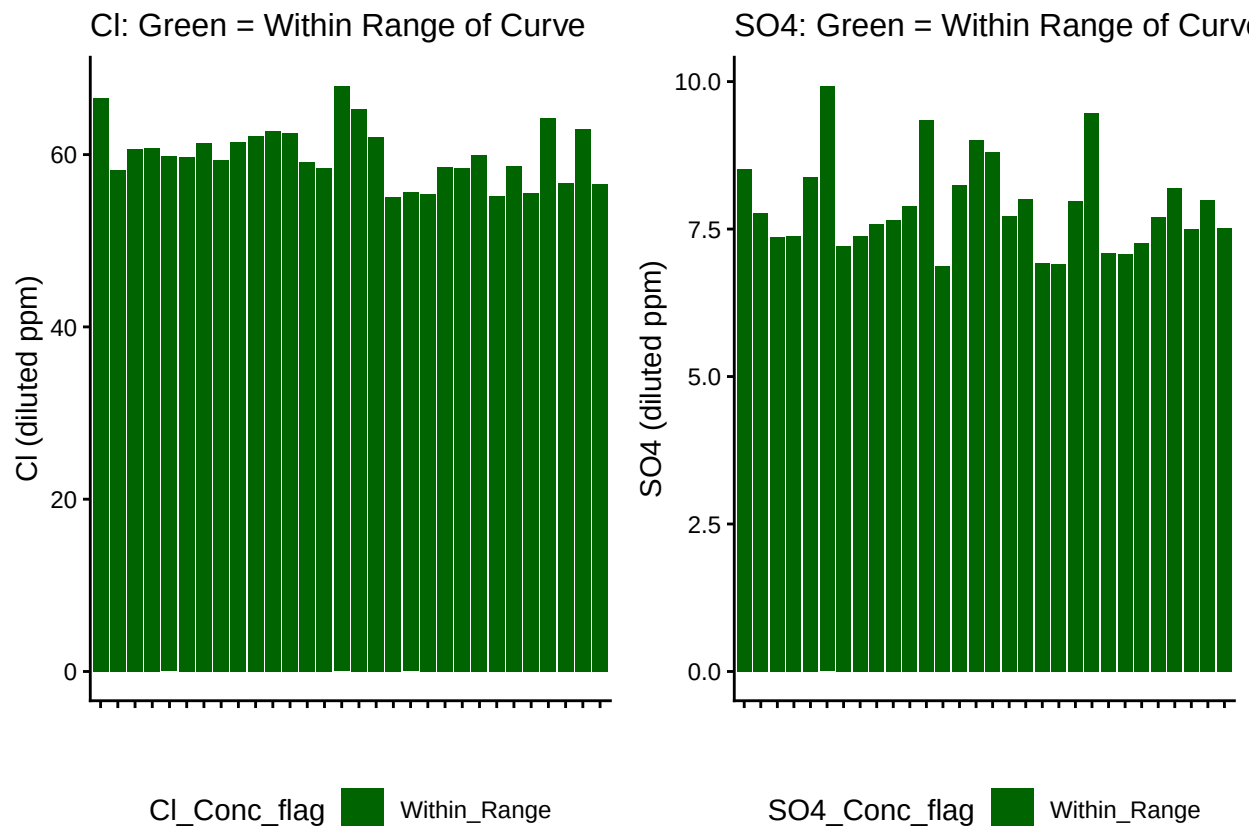


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

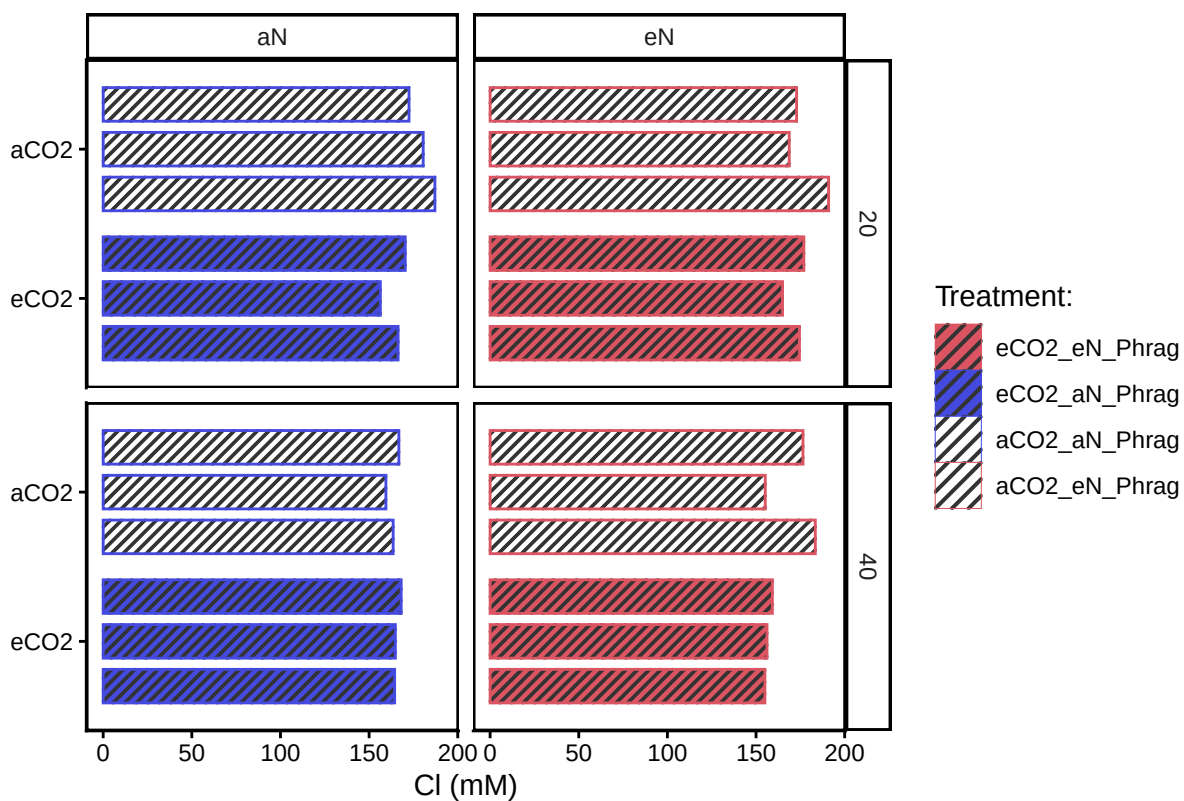
0.9 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```

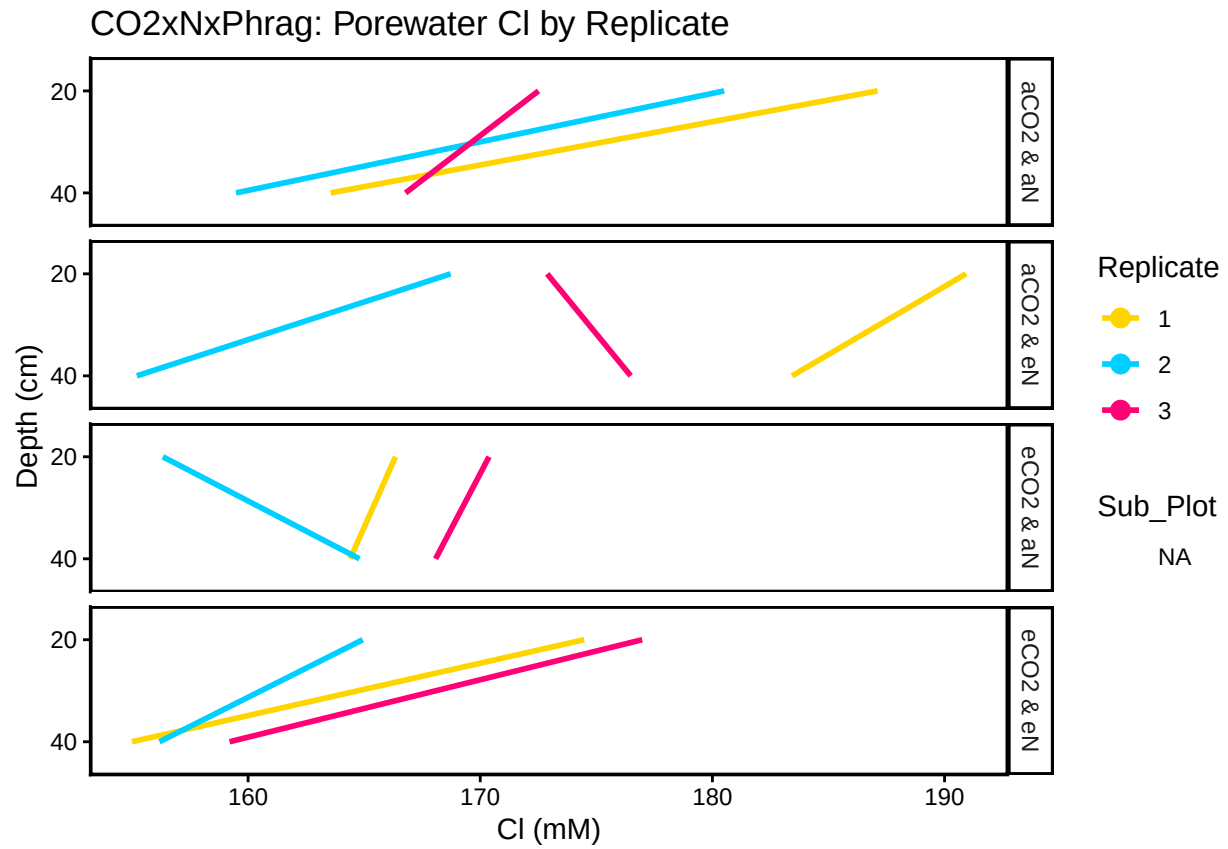
0.10 Visualize Data

```
## Visualize Data
```

CO2xNxPhrag: Porewater Chloride

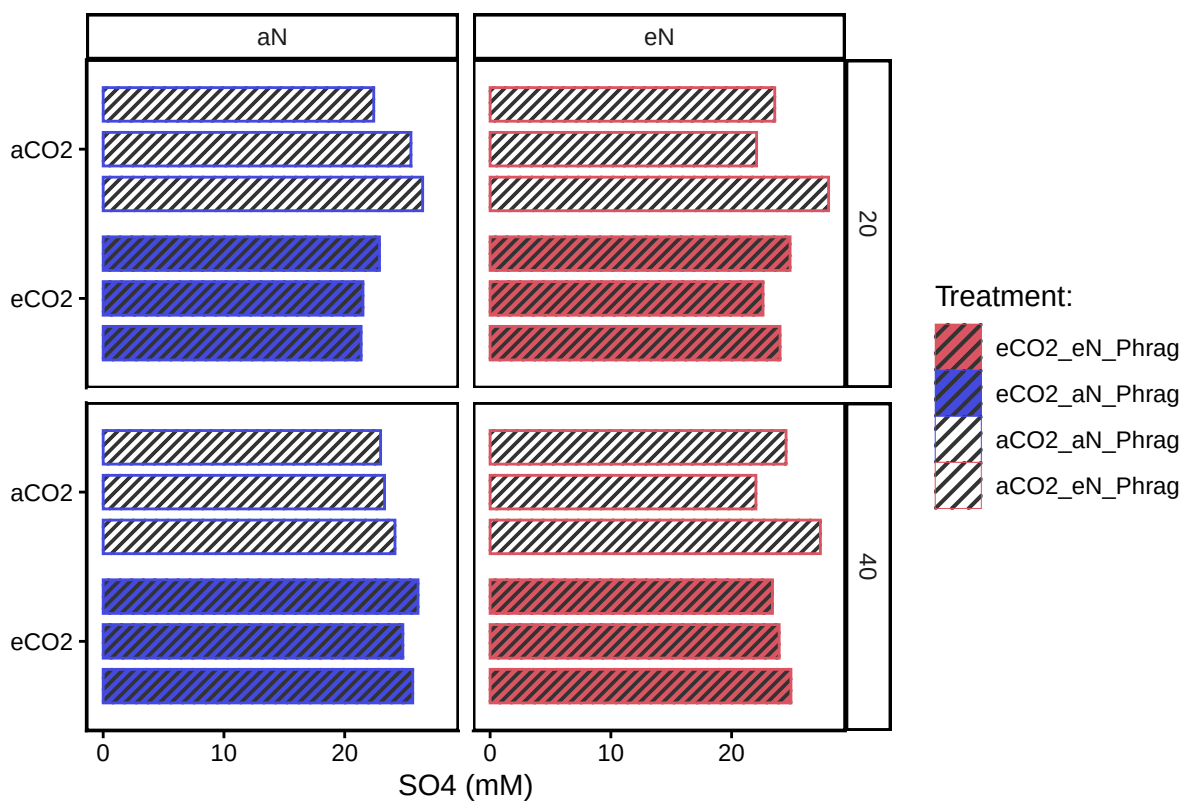


Warning: Removed 24 rows containing missing values or values outside the scale range
('geom_point()').

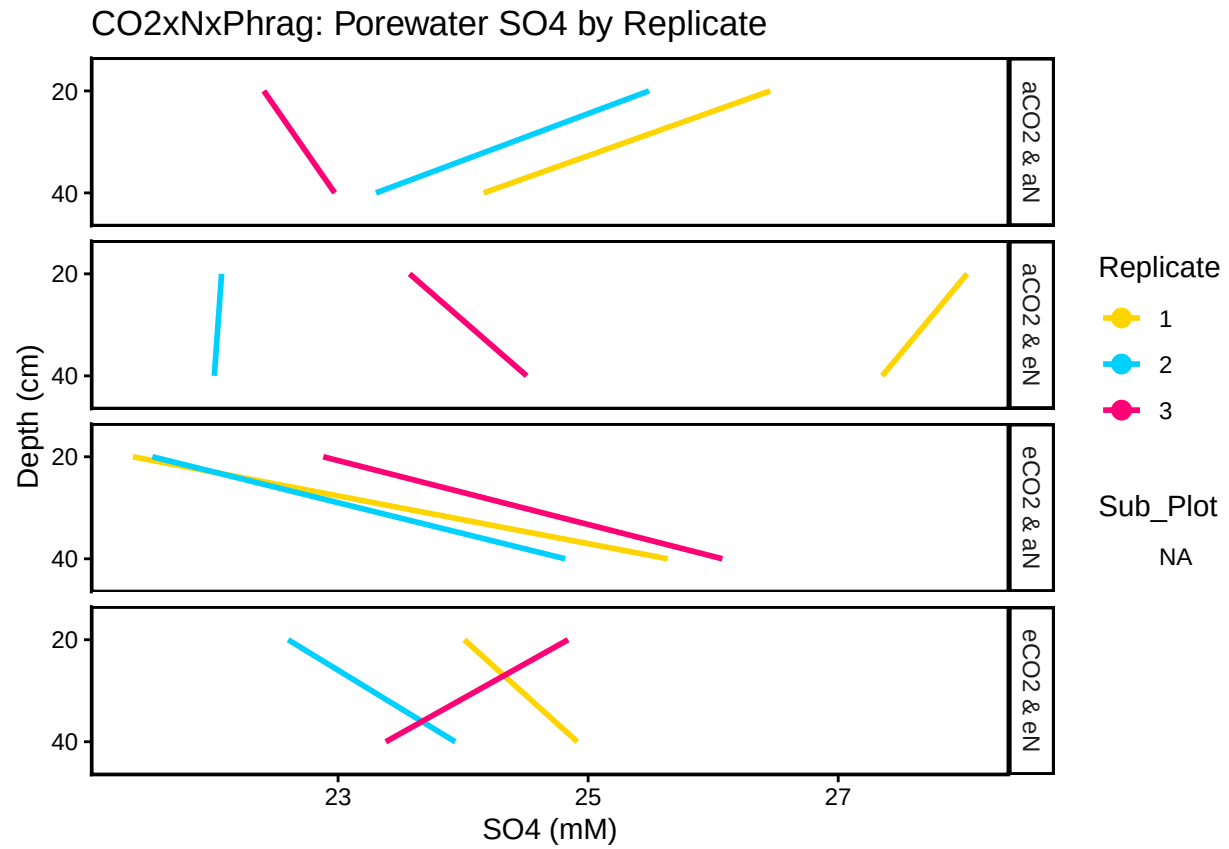


Visualize S04 Data

CO2xNxPhrag: Porewater Sulfate



Warning: Removed 24 rows containing missing values or values outside the scale range
('geom_point()').



0.11 Export Processed Data

#end